

A Palm-Through Interaction Technique for Controlling IoT Devices

Zhengchang Yang*
NAIST, Japan

Naoya Isoyama†
NAIST, Japan

Nobuchika Sakata‡
Ryukoku University, Japan

Kiyoshi Kiyokawa§
NAIST, Japan

ABSTRACT

In this study, we propose a palm-through interaction technique to control smart devices by using an augmented reality head-mounted display. The contributions are as follows: 1) The proposed method is intuitive to aim at and touch the remote object of concern, 2) natural haptic feedback is generated. The user study suggests that the palm-through interface is more intuitive because the user interface is similar to that of a smartphone, and brings users haptic confidence.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction techniques

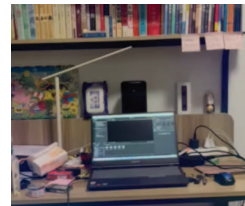
1 INTRODUCTION

The integration of augmented reality (AR) with Internet-of-Things (IoT) technology enables the creation of a link among the actual world, virtual objects, and sensor information from IoT devices. For example, head-mounted displays (HMDs) augment the human ability to experience the world from an egocentric perspective by sensing and showing the information which is imperceptible otherwise. Thus, they expand the information's richness and enable context-aware applications.

A typical way to communicate with IoT devices is by voice commands delivered via intelligent assistants. However, giving voice orders to spatially distributed IoT devices is difficult for the assistant to comprehend and execute, and it is susceptible to background noises. An alternative option is to communicate with IoT devices utilizing gestures. However, they can be ambiguous on their own as a single gesture command can be polysemous, e.g., volume-up or brightness-up for different devices. One strategy for resolving ambiguity is multimodal interaction, which commonly combines voice and gesture [1, 5]. These methods require additional sensing devices placed on the user's body or in the environment to determine the target device or its location of concern.

We introduce a palm-based see-through interface as a novel interaction technique to operate IoT devices, enabling remote interaction from afar. We use an optical see-through HMD as the interaction device. No extra sensors in the surrounding or on the hands are required. Thus, the device can become as simple as possible.

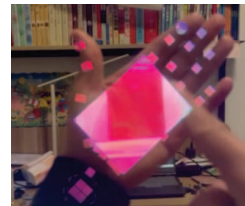
One reason for proposing this interface is to ensure that it is socially acceptable. According to several studies, users prefer a silent input style in specific scenarios [5]. For instance, in noisy surroundings, users prefer visual modality or gestures. Another motivation is that lacking haptic feedback presents another challenge when interacting with virtual objects in AR environments, because haptic feedback enhances the naturalness of an interaction. The primary selection approach utilized in modern AR devices is mid-air selection via touch of the virtual buttons. Without haptic feedback, users have a tough difficulty in confirming the selection. Previous



(a) User's view without any operations.



(b) Raising the hand to occlude the speaker, then the palm-through interface appears.



(c) The user selects the speaker by directly touching it.



(d) UI components appear on the fingers with which the user manipulates the device.

Figure 1: The proposed palm-through interface in action. Note that the reddish color is due to the HoloLens 2 optics, and the see-through view looks bigger than it is supposed to be due to the camera position far behind the eye. It appears better with the naked eyes.

research [3] has discovered that the tactile cues detected by the palm and the pointing finger enable users to efficiently select objects.

2 THE PALM-THROUGH INTERFACE

Inspired by [4], we follow the idea of using the user's own body as a surface for mobile interaction. In this article, we propose a novel interaction technique, a palm-based see-through interface, which serves as a selection method in 3D space. Compared to traditional 3D selection methods, such as raycasting, gaze, mid-air touch, and voice commands, the proposed approach provides haptic feedback and a silent way to manipulate devices.

The palm-through interface provides a selection method and an interaction to the users for controlling devices that are out of reach. The see-through property lets the user see the occluded view behind the palm, and the palm and fingers' abundant tactile features and natural divisions enable it to be used as interaction surfaces. Concretely, we can consider the palm-through interface as a touch-based device like a smartphone. As users are already familiar with how to use smartphones, the proposed interface is intuitive to them. The interaction principle of the interface is shown in Fig. 1

A former study [2] reported that touch accuracy on the palm is worse than it is on a touchscreen. The problem becomes more severe when the objects are remote and small. To mitigate this, the interface supports pinch to zoom in the image like smartphones for performing a more precise selection.

3 IMPLEMENTATION

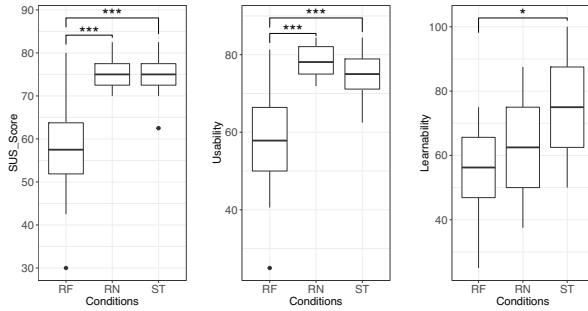
Our system was built using a HoloLens 2 optical see-through AR HMD with an additional camera attached. The reason of not using

*e-mail: yang.zhengchang.xr6@is.naist.jp

†e-mail: isoyama@is.naist.jp

‡e-mail: sakata@rins.ryukoku.ac.jp

§e-mail: kiyo@is.naist.jp



(a) Overall SUS score (b) Usability scores (c) Learnability scores
Figure 2: The result of SUS with significance annotated by stars per category (*: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$).

the built-in camera is that it can't achieve acceptable frame rate due to limited computation power and development kit limitations. Our system runs on a PC with an NVIDIA GPU and the palm-through view can achieve 16 fps under 720P resolution by using the webcam.

When the user raises the non-dominant hand, the hand would occlude the view we desired. To deal with this issue, we keep on taking a background image until the user raises the non-dominant hand, then we keep on taking another image that contains the user's hand. With these two images, finding the correspondences is done by matching the image features by using the ORB feature descriptor. Then, we warp the background image to match the image with hand. In order to find a better view alignment within the warped image, we involve hand joint tracking data acquired by the HoloLens 2.

4 EVALUATION

In the user study, we compared the palm-based see-through interface (ST) with two baselines, Raycasting Far (RF) and Raycasting Near (RN). RF uses a ray emitted from the user's hand to select remote IoT devices and interact with those remote UIs, and RN still uses a ray to select remote devices and mid-air touch to manipulate devices. We recruited 12 participants (average age 25.0, ranging from 20 to 32, 7 females) for the experiment. The order of the three conditions is randomized for each participant. Each participant carried out a fixed series of appliance interactions in each of the conditions, and fill out questionnaires.

SUS From Fig. 2, for SUS and usability scores, RN and ST obtain similar average scores which are much higher than that of RF. RN achieves slightly higher scores than ST. The result from the ANOVA analysis shows that a statistically significant difference (SSD) was found in SUS score ($F = 18.65$, $p < 0.001$), and a TukeyHSD test found SSDs between RN and RF ($p < 0.001$) and between ST and RF ($p < 0.001$), respectively. Another SSD was found in Usability ($F = 15.18$, $p < 0.001$), and a TukeyHSD test found SSDs between RN and RF ($p < 0.001$) and between ST and RF ($p < 0.001$), respectively. We also found an SSD ($F = 5.423$, $p < 0.01$) in Learnability, and a TukeyHSD test found an SSD between ST and RF ($p < 0.017$).

NASA-TLX RF obtained the highest average score in mentally demanding and physically demanding, RN is in the middle, and ST is at the least level. Regarding performance, RN and ST are similar but much better than RF in average score. For temporal demand, effort, and frustration, RF shows the worst result in average score, ST is in the middle, and RN is the best. However, The result from ANOVA analysis shows that there are no SSDs among the six questions, which means three scenarios are at a similar level of workload.

Customized Questionnaire In Customized Questionnaire, We measure two aspects of user's feelings of touch, the sense of touch ("I felt the sense of touch of remote objects") and the user's confidence of haptic feedback ("I felt confident when receiving haptic feedback") on a five-point Likert scale from "disagree" to "agree". From the Friedman test, we found an SSD ($p < 0.001$), a post-hoc test found SSDs between RF and ST ($p < 0.001$), and between RN and ST ($p < 0.05$), in both questions respectively.

Task Completion Time We also calculate the mean task completion time per participant and condition. The result from the Friedman test shows that all tasks yielded SSDs ($p < 0.01$). The average overall task completion time for ST is nearly the same as that of RN. RF is the most time-consuming method.

5 DISCUSSION

The general advantages of our approach are that it makes appliances easily accessible from a distance. Furthermore, our interaction approach was easy to learn over all scenarios, we believe this to be the result of the similarity of our interface to typical smartphone interfaces, and the user can easily understand how our approach works by transferring the skills. This makes the control very intuitive, as none of the interactions has to be remembered.

Despite the fact that we introduced a mechanism that allows the user to lower his hand to operate IoT devices after selection, we are still concerned that the user's arm fatigue would influence the usability of our interface, particularly after a prolonged time of interaction. However, from the results of NASA-TLX and Usability scores, we know that there is no SSD found on task load, which means all three conditions share similar levels of workload. It was observed that most of the participants would put their hands down to manipulate those devices, and they gave positive feedback on our interface.

One strong advantage the proposed approach has over the other interaction conditions, is that it is able to provide haptic feedback as exemplified by a participant's comment, "the haptic feedback is great, I feel more comfortable." This goes with the fact that the palm-through interface was preferred for touch sensation and haptic confidence from Customized Questionnaire.

6 CONCLUSION

We proposed a palm-through interaction technique to control smart devices by using an AR HMD and confirmed its effectiveness. The prototype system in the user study can only handle three simulated devices in a single office, however, the principle can be used to a much bigger number of devices and a much larger area. In the future, by employing hierarchical control, we extend one layer to multiple layers for more diverse control due to limited finger areas, the scalability of our interface could be much better.

REFERENCES

- [1] R. A. Bolt. "put-that-there" voice and gesture at the graphics interface. In *Proceedings of the 7th annual conference on Computer graphics and interactive techniques*, pp. 262–270, 1980.
- [2] S. Gustafson, C. Holz, and P. Baudisch. Imaginary phone: learning imaginary interfaces by transferring spatial memory from a familiar device. In *Proceedings of the 24th annual ACM symposium on User interface software and technology*, pp. 283–292, 2011.
- [3] S. G. Gustafson, B. Rabe, and P. M. Baudisch. Understanding palm-based imaginary interfaces: the role of visual and tactile cues when browsing. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 889–898, 2013.
- [4] C. Harrison, D. Tan, and D. Morris. Skinput: appropriating the body as an input surface. In *Proceedings of the SIGCHI conference on human factors in computing systems*, pp. 453–462, 2010.
- [5] R. Kang, A. Guo, G. Laput, Y. Li, and X. Chen. Minuet: Multimodal interaction with an internet of things. In *Symposium on Spatial User Interaction*, pp. 1–10, 2019.